Simulation Analytics — Metric definitions

Metric	Definition	Formula / source
Sheets overview		
Incoming Trains	One row per inbound train, full lifecycle on a single line (identity, timestamps, derived waits/durations).	tab 2; built by build_incoming_trains_sheet.
Wagon Groups	One row per wagon group, full lifecycle (parent train → classification → parent OBT → departure).	tab 3; built by build_wagon_groups_sheet.
Outbound Trains	One row per outbound train (OBT), full lifecycle on the classification side (OBTP, exit gate, DEPD).	tab 4; built by build_outbound_trains_sheet.
WG Yard Times	Per-WG cross-check of arrival-yard and classification-yard dwell times, with all the supporting timestamps shown.	built by build_wg_yard_times_sheet.
Activities (raw)	One row per completed activity instance. Identity (type, id, entity, length), drive context (from/to/distance), timestamps (initialized/started/completed), duration.	built by build_activities_raw_sheet.
OBTs (raw) / WGs (raw)	Flat rows for outbound trains and classified wagon groups; redundant with the per-entity sheets but kept for legacy cross-references.	built by build_obts_raw_sheet / build_wgs_raw_sheet.
Incoming Trains KPIs		
Entry Queue wait (hh:mm:ss)	Time the train sat in the entry queue waiting for an arrival track to free up. Pure queue time — does NOT include the drive.	Incoming Trains!M = (E − D) = ArrivalDriveStart − EntryToStation
Arrival drive duration (hh:mm:ss)	Time the train spent physically driving from the entry gate to its assigned arrival track.	Incoming Trains!N = (F − E) = AtArrivalTrack − ArrivalDriveStart
ITP wait (hh:mm:ss)	Time between ITP being submitted to ResourceCU and the work actually commencing. Captures worker + shunt-loco allocation + travel-to-site time.	Incoming Trains!O = (H − G) = ITP_Starts − ITP_Init
ITP duration (hh:mm:ss)	Time the ITP work itself ran, once resources arrived.	Incoming Trains!P = (I − H) = ITP_Ends − ITP_Starts
PushOff wait (hh:mm:ss)	Time between PushOff being submitted and the work actually commencing. Captures worker re-allocation + passage-track wait.	Incoming Trains!Q = (K − J) = PushOff_Starts − PushOff_Init
PushOff duration (hh:mm:ss)	Time the entire PushOff envelope ran (all sub-drives + loco pull-backs between them).	Incoming Trains!R = (L − K) = PushOff_Ends − PushOff_Starts
Total in arrival yard (hh:mm:ss)	Total time the inbound train physically occupied an arrival track (from arrival to last WG pushed off).	Incoming Trains!S = (L − F) = PushOff_Ends − AtArrivalTrack
Total in system (hh:mm:ss)	End-to-end inbound time: from entry-queue arrival until PushOff completed.	Incoming Trains!T = (L − D) = PushOff_Ends − EntryToStation
Wagon Groups KPIs		
Time in arrival yard (hh:mm:ss)	Time the WG was on the arrival track as part of its parent train, from arrival to being pushed off.	Wagon Groups!M = (G − E) = PushOffDriveEnds − AtArrivalTrack
Time in classification yard (hh:mm:ss)	Time the WG sat on its classification track, from arriving there (= PushOffDrive ends) until the parent OBT physically departed.	Wagon Groups!N = (L − G) = DepartureDriveEnds − PushOffDriveEnds
Idle time on classification track (hh:mm:ss)	Time the WG waits after getting coupled or secured until the OBTP activity starts — the 'doing nothing' window after its own prep is done but before OBT-wide work begins.	Wagon Groups!P = (I − O) = OBTP_Starts − SEC/COP_End
Classification context	Plain-language narrative of the WG's time on the classification track: how it joined (secured first, or coupled to the sibling already there), the wait before each later sibling coupled, the OBTP window, and departure.	Wagon Groups!Q; built in parse_wagon_groups.
Outbound Trains KPIs		
OBTP wait (hh:mm:ss)	Time between OBTP being submitted and actually starting. Captures worker + train-loco allocation + travel.	Outbound Trains!K = (F − E) = OBTP_Starts − OBTP_Init
OBTP duration (hh:mm:ss)	Time the OBTP work itself ran.	Outbound Trains!L = (G − F) = OBTP_Ends − OBTP_Starts
DEPD gate wait (hh:mm:ss)	Time the OBT was ready to depart (DEPD submitted) but stuck waiting for the single exit-gate resource. Headline bottleneck metric given exitGateCount = 1.	Outbound Trains!M = (I − H) = DEPD_Starts − DEPD_Init
DEPD duration (hh:mm:ss)	Time the train spent physically driving from its classification track to the exit gate.	Outbound Trains!N = (J − I) = DEPD_Ends − DEPD_Starts
Total on classification track (hh:mm:ss)	End-to-end outbound time, from OBT-creation moment (≈ OBTP init) until DEPD completes.	Outbound Trains!O = (J − E) = DEPD_Ends − OBTP_Init
Activities (raw) reference		
Activity row	One row per completed activity (Started AND Completed present). Pending / cancelled activities are excluded from this sheet.	parse_activity_durations in analytics.py.
From / To / Drive length	Populated only for the three driving activity types (ArrivalDrive, DepartureDrive, PushOffDrive). Distance values come from static maps near the top of analytics.py (ported from C# Models/*DriveActivity.cs and Models/PushOffActivity.cs).	ARRIVAL_DRIVE_DISTANCES + DEPARTURE_DRIVE_DISTANCES + PUSH_ARRIVAL_LEG + PUSH_CLASS_LEG (in analytics.py).
Duration (hh:mm:ss)	Wall-clock time the activity spent in its 'running' state. Rendered as hh:mm:ss across the whole workbook.	Activities (raw)!K = Completed − Started

Metric	Definition	Formula / source
Activity type glossary		
Activity type abbreviations	ARRD = ArrivalDrive (train enters yard on own loco); ITP = IncomingTrainPreparation (workers prep arriving train); PO = PushOff (parent envelope); POD = PushOffDrive (one sub-drive per consecutive-same-destination cut of WGs); SEC = Securing (first WG on a classification track); COP = Coupling (joining WG to existing WGs on track); OBTP = OutboundTrainPreparation (workers + train loco prep OBT); DEPD = DepartureDrive (OBT exits through gate).	(reference — see Models/Activity.cs GetActivityAbbreviation).
WG → OBT matching		
How parent OBT is derived for a WG	WGs are matched to OBTs by classification track + time window: an OBT created at time T on track X owns every WG that arrived on track X at or before T and wasn't already claimed by an earlier OBT on the same track. Greedy, earliest-OBT-first.	parse_obts_and_wgs in analytics.py.